It is recommended that you spend about 30 minutes on this question.

2 Many musical instruments contain wires which are stretched between two fixed supports. These wires are often made to resonate in their fundamental mode of vibration, as shown in Fig. 2.1.

Fig. 2.1

The frequency of the note produced by the wire when it vibrates depends on

(i) the length of the wire between the supports,

(ii) the tension in the wire,

(iii) the mass per unit length of the wire.

Design an experiment to investigate how the resonant frequency of a wire vibrating in its fundamental mode depends on the tension in the wire.

You should draw a diagram showing the arrangement of your equipment. In your account you should pay particular attention to

(a) the method by which the wire would be made to vibrate at a known frequency,

(b) how the resonant frequency and the tension would be measured,

(c) the procedure to be followed,

(d) control of variables,

(e) any safety precautions which you would take.

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